

6	(a) charge = current $\times$ time	B1	[1]
	(b) (i) $P = V^2 / R$ $= (240)^2 / 18 = 3200 \text{ W}$	C1 A1	[2]
	(ii) $I = V / R = 240 / 18 = 13.3 \text{ A}$	A1	[1]
	(iii) charge = $It = 13.3 \times 2.6 \times 10^6$ $= 3.47 \times 10^7 \text{ C}$	C1 A1	[2]
	(iv) number of electrons = $3.47 \times 10^7 / 1.6 \times 10^{-19} (= 2.17 \times 10^{26})$ number of electrons per second = $2.17 \times 10^{26} / 2.6 \times 10^6 = 8.35 \times 10^{19}$	C1 A1	[2]

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